Hoarding Management System (HMS)

Complete User Manual

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1. System Overview

The **Hoarding Management System (HMS)** is a full-stack enterprise application built to manage the complete lifecycle of outdoor advertising holdings (hoardings / billboards). It covers:

- **Inventory** - Digital register of all hoarding sites
- **Commercial** - Booking, billing, and client management
- **Operations** - Field tasks, inspections, and maintenance
- **Finance** - Fully integrated double-entry accounting with auto-posted journal entries

Core Principle: Every financial transaction (invoice raised, payment received, vendor payment made) automatically generates a balanced journal entry in the ledger no manual bookkeeping required.

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2. User Roles and Access

Role	Access Level
Admin	Full access to all modules including accounting, user management, and reports
Staff	Access to operational modules tasks, inspections, location suggestions, task execution

- Login is required for all protected routes.
- Staff cannot access the accounting module or admin settings.
- All task executions require GPS verification (latitude/longitude captured at the time of marking a task complete).

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3. Master Data Setup

Master data is the foundation of the system. It must be configured **before** any holdings, bookings, or billing can be created.

3.1 Cities

Manages the geographic locations where holdings are situated.

Field	Description
Name	City name (e.g., Indore)
State	State name (e.g., Madhya Pradesh)
Country	Default: India
Active	Toggle to activate/deactivate

Unique Rule: No two cities with the same name+state combination are allowed.

3.2 Holding Types

Classifies the type of hoarding structure.

Field	Description
Name	E.g., Unipole, Gantry, Wall Mount, LED Board
Description	Optional description

3.3 HSN Codes

HSN (Harmonized System of Nomenclature) codes are used for GST compliance on invoices.

Field	Description
Code	4/6/8-digit HSN code per GST rules
Description	What it covers
GST Rate (%)	Default GST rate applied to this code

Each holding is tagged with an HSN code. When an invoice is generated, the HSN code from the booking flows into the invoice automatically.

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4. Holdings Management

A **Holding** is the core asset of the system – a physical hoarding/billboard site.

4.1 Holding Fields

Field	Description
Code	Unique identifier (e.g., IND-001)
Name	Descriptive name
Address	Physical address
Latitude / Longitude	GPS coordinates for map location
Width x Height	Physical dimensions in feet/meters
Total Area	Computed display area (sq. ft.)
Illumination	LIT / NON_LIT / DIGITAL
Facing	Road-facing direction
Landmark	Nearest landmark
Status	AVAILABLE / BOOKED / UNDER_MAINTENANCE / INACTIVE
Maintenance Cycle	Days between scheduled maintenance (default: 90 days)
City	Linked from master data
Holding Type	Linked from master data
HSN Code	GST classification from master data

4.2 Holding Status Flow

```
AVAILABLE  -->  BOOKED  -->  COMPLETED  -->  AVAILABLE
                   |
             UNDER_MAINTENANCE  -->  AVAILABLE / INACTIVE
```

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5. Ownership Contracts

Every holding has an **Ownership Contract** that records the legal/commercial arrangement with the site owner (landlord).

5.1 Contract Fields

Field	Description
Contract Number	Unique contract reference
Owner Name	Name of the site owner/landlord
Owner Type	GOVERNMENT / MUNICIPAL / VILLAGE_PANCHAYAT / PRIVATE
Owner Contact / Email / Address	Owner communication details
Rent Amount	Monthly/periodic rent to be paid
Rent Cycle	MONTHLY / QUARTERLY / HALF_YEARLY / YEARLY
Start Date / End Date	Contract validity period
Security Deposit	One-time refundable deposit
Status	ACTIVE / EXPIRED / TERMINATED / PENDING
Holding	The holding site this contract belongs to

A holding can have multiple contracts (e.g., old expired + new active), but typically one active contract at a time. Ownership contracts are also linked to **Vendors** ⬩ the vendor record tracks the payee for rental payments.

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6. Client Management

Clients are the advertising companies or brands that book holdings.

6.1 Client Fields

Field	Description
Name	Company/client name
Contact Person	Primary contact
Email / Phone	Communication details
GST Number	GST registration (for GST invoices)
PAN Number	PAN for compliance
Address	Billing address
City	Linked city
Ledger	Auto-assigned Accounts Receivable ledger

Ledger Auto-Creation: When the first invoice for a client is posted, the system automatically creates a sub-ledger under "Accounts Receivable" for that client (e.g., `ABC Pvt Ltd - Receivable`). This ledger tracks all outstanding amounts for the client.

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7. Vendor Management

Vendors are site owners / landlords to whom rental payments are made.

7.1 Vendor Fields

Field	Description
Name	Vendor/owner name
Contact Person / Email / Phone	Communication details
GST Number / PAN Number	Tax compliance
Address	Vendor address
City	Linked city
Ownership Contract	Linked contract (optional)
Ledger	Accounts Payable ledger (required)

Ledger Auto-Creation: If a vendor payable ledger does not exist, the system creates a sub-ledger under "Accounts Payable" automatically (e.g., `Ram Landlord - Payable`) when the first payment is processed.

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8. Bookings

A **Booking** is the commercial agreement between HMS and a Client to display advertising on a holding for a defined period.

8.1 Booking Fields

Field	Description
Booking Number	Unique reference (e.g., BK-2024-00001)
Client	The advertiser
Holding	The hoarding site
Start Date / End Date	Campaign period
Monthly Rate	Rate charged per month
Total Amount	Monthly Rate x duration
Billing Cycle	MONTHLY / QUARTERLY / HALF_YEARLY / YEARLY
Free Mountings	Number of free mounting services included
Total Mountings	Total mounting visits planned
Status	CONFIRMED / ACTIVE / COMPLETED / CANCELLED

8.2 Booking to Invoice Link

```
Booking  -->  Invoice(s)  -->  Receipt(s)
```

- A booking can have multiple invoices (one per billing cycle)
- Each invoice can have multiple receipts (partial payments allowed)

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9. Advertisements

An **Advertisement** is the creative/campaign that runs within a booking period.

9.1 Advertisement Fields

Field	Description
Campaign Name	Name of the advertising campaign
Brand Name	Brand being advertised
Artwork Description	Notes on the creative
Artwork URL	Link to artwork file
Installation Date	When artwork is installed
Removal Date	When artwork is removed
Status	PENDING / INSTALLED / ACTIVE / REMOVED / COMPLETED
Booking	Parent booking this advertisement belongs to

Each advertisement can have Tasks assigned to it (e.g., INSTALLATION, MOUNTING tasks).

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10. Tasks and Field Operations

The **Task** module manages all field work – installations, mountings, maintenance, and inspections assigned to field staff.

10.1 Task Fields

Field	Description
Title	Task name
Task Type	INSTALLATION / MOUNTING / MAINTENANCE / INSPECTION
Priority	LOW / MEDIUM / HIGH / URGENT
Status	PENDING / IN_PROGRESS / UNDER_REVIEW / COMPLETED / CANCELLED
Scheduled Date	When the task should be performed
Completed Date	Actual completion date
Assigned To	Staff user responsible
Estimated Cost / Actual Cost	Budget vs. actual
Holding	Linked holding site
Advertisement	Linked advertisement (if applicable)

10.2 Task Execution (Field Completion)

When a staff member completes a task in the field, they submit a Task Execution record with proof:

Field	Description
Status	COMPLETED / UNDER_REVIEW / CANCELLED
Condition	EXCELLENT / GOOD / FAIR / POOR / CRITICAL
Remarks	Field notes
GPS Coordinates	Latitude + Longitude (captured automatically from device)
Front View Photo	Mandatory photo from front
Left View Photo	Mandatory photo from left
Right View Photo	Mandatory photo from right

GPS Verification: The system captures the field staff GPS coordinates at the time of task execution to verify on-site presence. Photos are uploaded and stored securely.

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11. Inspections and Maintenance

11.1 Inspections

Regular condition checks of holdings are recorded as Inspections.

Field	Description
Inspection Date	Date of inspection
Inspector Name	Name of the inspector
Condition	EXCELLENT / GOOD / FAIR / POOR / CRITICAL
Illumination OK	Whether lighting is working
Structure OK	Whether structure is intact
Visibility OK	Whether visibility is unobstructed
Remarks	Inspector notes
Photos	Multiple inspection photos attached

11.2 Maintenance Records

Field	Description
Maintenance Type	PAINTING / STRUCTURAL_REPAIR / ELECTRICAL / CLEANING / REPLACEMENT / OTHER
Description	What was done
Cost	Amount spent
Performed Date	Date of maintenance
Performed By	Name of service provider
Status	SCHEDULED / IN_PROGRESS / COMPLETED / CANCELLED

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12. Billing Invoices

An **Invoice** is the formal tax document raised to a client for hoarding rental services.

12.1 Invoice Fields

Field	Description
Invoice Number	Unique reference (system-generated)
Invoice Date	Date of invoice
Due Date	Payment due date
Client	Billed client
Booking	Related booking
HSN Code	GST classification
Subtotal	Amount before tax
CGST Rate / Amount	Central GST (e.g., 9%)
SGST Rate / Amount	State GST (e.g., 9%)
IGST Rate / Amount	Integrated GST for inter-state (e.g., 18%)
Total Amount	Subtotal + All taxes
Paid Amount	Amount received so far
Status	DRAFT / SENT / PAID / PARTIALLY_PAID / OVERDUE / CANCELLED

12.2 Invoice Line Items

Field	Description
Description	Service description
HSN Code	Per-item HSN code
Quantity	Number of units (months/periods)
Rate	Rate per unit
Amount	Quantity x Rate
GST Rate / Amount	Applicable GST
Total	Amount + GST Amount

When an invoice is posted, the system automatically creates a Journal Entry (see Section 15.3).

13. Billing Receipts Client Payments

A **Receipt** records money received from a client against an invoice.

13.1 Receipt Fields

Field	Description
Receipt Number	Unique reference
Receipt Date	Date of payment receipt
Amount	Amount received
Payment Mode	CASH / CHEQUE / NEFT / RTGS / UPI / CARD / OTHER
Reference No	Cheque number / UTR / transaction ID
Client	Who paid
Invoice	Which invoice this is against
Cash/Bank Ledger	Which Cash or Bank account received the money
Notes	Additional remarks

When a receipt is saved, the system automatically creates a Journal Entry and updates the invoice paid amount. If paid amount equals total amount, invoice becomes PAID; otherwise PARTIALLY_PAID.

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14. Vendor Payments

A **Payment** records money paid out to a vendor (site owner/landlord) for rent.

14.1 Payment Fields

Field	Description
Payment Number	Unique reference (e.g., PAY-2024-00001)
Payment Date	Date of payment
Amount	Amount paid
Payment Mode	CASH / CHEQUE / NEFT / RTGS / UPI / CARD / OTHER
Reference No	Transaction reference
Vendor	Who was paid
Cash/Bank Ledger	Account from which payment was made
Notes	Remarks

When a payment is saved, the system automatically creates a Journal Entry (see Section 15.5).

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15. Accounting Module

HMS has a **fully integrated double-entry accounting system**. Every billing and payment action automatically posts a balanced journal entry.

15.1 Chart of Accounts (Ledgers)

Ledger Types

Type	Examples
ASSET	Cash in Hand, Bank Account, Accounts Receivable
LIABILITY	Accounts Payable, GST Payable
INCOME	Hoarding Revenue / Rental Income
EXPENSE	Rent Expense, Maintenance Expense
EQUITY	Capital, Retained Earnings

Ledger Flags (Special Purpose Markers)

Flag	Purpose
isCash	Marks "Cash in Hand" ⬆ used for cash receipts/payments
isBank	Marks a "Bank Account" ⬆ used for bank receipts/payments
isReceivable	Marks parent group for Client Receivable sub-ledgers
isPayable	Marks parent group for Vendor Payable sub-ledgers
isRevenue	Marks the Revenue ledger for invoice auto-posting
isTaxOutput	Marks GST output ledgers (CGST, SGST, IGST Payable)
isTaxInput	Marks GST input ledgers (for purchase invoices)
isGroup	Group/header ledger ⬆ child ledgers hold actual entries

Recommended Initial Ledger Setup

```
Assets
  Cash in Hand              [isCash]
  HDFC Bank Current A/c     [isBank]
  ICICI Bank A/c            [isBank]
  Accounts Receivable       [isReceivable, isGroup]
    ABC Pvt Ltd - Receivable    <- auto-created on first invoice
    XYZ Corp - Receivable       <- auto-created on first invoice

Liabilities
  CGST Payable              [isTaxOutput]
  SGST Payable              [isTaxOutput]
  IGST Payable              [isTaxOutput]
  Accounts Payable          [isPayable, isGroup]
    Ram Landlord - Payable      <- auto-created on first payment
    City Corp - Payable         <- auto-created on first payment

Income
  Hoarding Rental Revenue   [isRevenue]

Expense
  Site Rent Expense
  Maintenance Expense
```

15.2 Journal Entries How They Are Auto-Created

Every time a financial document is saved (Invoice / Receipt / Payment), the system:

1. Looks up the relevant ledgers using the flag system
2. Builds a balanced set of debit and credit lines
3. Validates that Total Debits = Total Credits (double-entry rule)
4. Creates a JournalEntry with status POSTED
5. Links the journal entry back to the source document
6. Assigns a unique journal number: JE-YYYY-NNNNN

Manual journal entries can also be created by Admins from Accounting > Journal Entries.

15.3 Invoice Journal Entry

Trigger: When an invoice is created/posted for a client booking.

What it represents: HMS has earned revenue and the client owes money.

#	Account	Debit (Dr)	Credit (Cr)
1	Client Receivable (e.g., ABC Pvt Ltd - Receivable)	Total Invoice Amount	
2	Hoarding Rental Revenue		Subtotal (before tax)
3	CGST Payable		CGST Amount
4	SGST Payable		SGST Amount
5	IGST Payable (if inter-state)		IGST Amount

Example Invoice of Rs 1,18,000 (Rs 1,00,000 + 9% CGST + 9% SGST):

Account	Dr	Cr
ABC Pvt Ltd - Receivable	Rs 1,18,000	
Hoarding Rental Revenue		Rs 1,00,000
CGST Payable		Rs 9,000
SGST Payable		Rs 9,000
Total	**Rs 1,18,000**	**Rs 1,18,000**

Effect on Books:

- Client receivable balance increases (they owe you more)
- Revenue increases
- GST liability to government increases

15.4 Receipt Journal Entry

Trigger: When a receipt is recorded (client pays an invoice).

What it represents: Cash/bank balance increases and client outstanding debt decreases.

#	Account	Debit (Dr)	Credit (Cr)
1	Cash / Bank Ledger (selected at receipt)	Receipt Amount	◆
2	Client Receivable (auto-fetched from client ledger)	◆	Receipt Amount

Example ◆ Client pays Rs 1,18,000 via NEFT into HDFC Bank:

Account	Dr	Cr
HDFC Bank Current A/c	Rs 1,18,000	◆
ABC Pvt Ltd - Receivable	◆	Rs 1,18,000
Total	**Rs 1,18,000**	**Rs 1,18,000**

Effect on Books:

- Bank balance increases by receipt amount
- Client receivable balance decreases (they owe you less)
- Invoice status updated to PAID or PARTIALLY_PAID

15.5 Vendor Payment Journal Entry

Trigger: When a payment is made to a vendor (landlord rent payment).

What it represents: Cash/bank decreases and vendor liability is cleared.

#	Account	Debit (Dr)	Credit (Cr)
1	Vendor Payable (auto-fetched from vendor ledger)	Payment Amount	◆
2	Cash / Bank Ledger (selected at payment)	◆	Payment Amount

Example ◆ Paying Rs 25,000 rent to Ram Landlord via NEFT from HDFC Bank:

Account	Dr	Cr
Ram Landlord - Payable	Rs 25,000	◆
HDFC Bank Current A/c	◆	Rs 25,000
Total	**Rs 25,000**	**Rs 25,000**

Effect on Books:

- Vendor payable balance decreases (liability cleared)
- Bank balance decreases by paid amount

15.6 Manual Journal Entries

Admins can create manual journal entries for:

- Opening balances
- Corrections and adjustments
- Depreciation and write-offs
- Expenses not covered by vendor payments

Validation Rules:

- Minimum 2 lines per entry
- Each line must have debit OR credit (not both)
- Total Debits must equal Total Credits (enforced by system)
- Status: DRAFT -> POSTED -> VOID

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16. Location Suggestions

Field staff can submit **Location Suggestions** ◆ proposed new hoarding sites discovered during field visits.

Field	Description
Address	Physical address of the proposed site
City	City where it is located
Landmark	Nearest landmark
GPS Coordinates	Latitude / Longitude
Description	Details about the site
Photos	Multiple site photos
Proposed Rent	Estimated rent/cost
Owner Name / Phone	Site owner contact
Status	PENDING / ACCEPTED / REJECTED

Admin reviews suggestions and either accepts (converts to a holding) or rejects them.

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17. Reports Module

Report	Description
Revenue Report	Total revenue by period, client, or holding
Occupancy Report	Which holdings are booked vs. available
Client Outstanding	Pending receivables per client
Vendor Payables	Pending payments to vendors
Booking Summary	All bookings with status and value
Task Summary	Field task completion rates
Maintenance Cost Report	Spending on maintenance by holding/type
Ledger Balance	Current balance of any ledger account

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18. System Flow End to End

```
STEP 1 – MASTER SETUP
  Add City -> Add Holding Type -> Add HSN Code

STEP 2 – SITE ACQUISITION
  Create Holding -> Create Ownership Contract -> Create Vendor (landlord)
  (Vendor Payable Ledger is auto-created)

STEP 3 – CLIENT ONBOARDING
  Create Client
  (Client Receivable Ledger auto-created on first invoice)

STEP 4 – BOOKING
  Create Booking (Client + Holding + Period + Rate)
  Holding status changes to BOOKED

STEP 5 – ADVERTISEMENT
  Create Advertisement under Booking
  Assign Installation / Mounting Tasks to field staff

STEP 6 – FIELD OPERATIONS
  Staff logs in, sees assigned Tasks
  Staff executes Task -> Submits GPS + 3 Photos
  Task status -> COMPLETED / UNDER_REVIEW

STEP 7 – BILLING (INCOME SIDE)
  Create Invoice for Booking period
  Journal Entry AUTO-POSTED:
    Dr: Client Receivable
    Cr: Revenue + CGST Payable + SGST Payable

  Record Receipt when client pays
  Journal Entry AUTO-POSTED:
    Dr: Bank / Cash
    Cr: Client Receivable
  Invoice status -> PAID / PARTIALLY_PAID

STEP 8 – VENDOR PAYMENT (EXPENSE SIDE)
  Record Payment to Vendor (monthly rent)
  Journal Entry AUTO-POSTED:
    Dr: Vendor Payable
    Cr: Bank / Cash
```

```
STEP 9 ⓪ REPORTING
  View outstanding balances, revenue, occupancy, task stats
```

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19. Module Relationship Map

```
City ------------------------------------------------+
                                                      |
HoldingType --------+                                 |
                    v                                 v
HsnCode -------> Holding <-- OwnershipContract <-- Vendor
                    |                                 |
                    v                                 v
Client -------> Booking                           Payment
  |                 |                                 |
  v                 v                                 v
Ledger(AR)    Advertisement                    JournalEntry
  |                 |
  v                 v
Invoice <---    Task --> TaskExecution (GPS + Photos)
  |
  v
Receipt --> JournalEntry --> JournalLines (Dr/Cr)
  |
Ledger (Cash/Bank)
```

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20. Glossary

Term	Definition
Holding	A physical hoarding/billboard site managed by HMS
Booking	Commercial agreement to display ads on a holding for a period
Advertisement	A specific campaign/creative running under a booking
Ownership Contract	Agreement with the site land/space owner
Vendor	A landlord or space owner to whom rent is paid
Client	An advertiser or agency that books holdings
Invoice	Tax document raised to a client for services
Receipt	Record of money received from a client
Payment	Record of money paid out to a vendor
Journal Entry	Accounting record with balanced debit/credit lines
Ledger	An account in the Chart of Accounts
HSN Code	Harmonized System of Nomenclature for GST classification
CGST	Central Goods and Services Tax
SGST	State Goods and Services Tax
IGST	Integrated GST for inter-state transactions
Dr (Debit)	Left side of a journal entry
Cr (Credit)	Right side of a journal entry
Accounts Receivable	Money owed to the company by clients
Accounts Payable	Money owed by the company to vendors
Task Execution	On-site completion record with GPS and photos
Location Suggestion	A field staff proposal for a new hoarding site

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